

Fig. 2: The IC datasheet pinouts

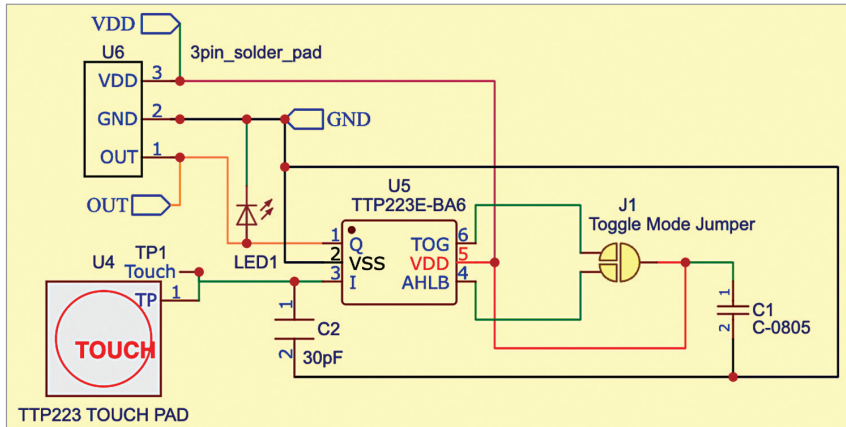
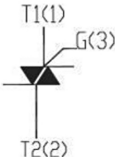


Fig. 3: Touch sensing circuit

Features

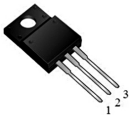
- High current triac
- Low thermal resistance with clip bonding
- High commutation (4 quadrant) or very high commutation (3 quadrant) capability



VOLTAGE RANGE
CURRENT

600/800 Volts
4 Ampere


TO-220AB


ITO-220AB


TO-252

MAXIMUM RATINGS AND ELECTRICAL CHARACTERISTICS

ELECTRICAL CHARACTERISTICS (Tj = 25°C, unless otherwise specified)

Symbol	Parameter	Conditions	Ratings	Unit
VDRM VRRM	Repetitive Peak Off-State Voltage	BT136-600	600	V
		BT136-800	800	
IT(RMS)	R.M.S On-State Current	Tc=110°C	4	A
ITSM	Surge On-State Current	tp=16.7ms/tp=10ms	25/27	A
I _p t	I _p t for fusing	Tp=10ms	3.1	A²s
PG(AV)	Average Gate Power Dissipation	Tj=125°C	1	W
IGM	Peak Gate Current	tp=20us Tj=125°C	2	A
Tj	Operating Junction Temperature		~40~125	°C
TSTG	Storage Temperature		~40~150	°C

Fig. 4: BT136 datasheet (Courtesy: IC datasheet)

SSR, capable of switching 100-250V AC loads (up to 4A, ~880W), suitable for bulbs, fans, and appliances. The components required to assemble this system are listed in the Bill of Materials table.

The system is divided into two main circuits: the touch sensing circuit and the solid-state relay (SSR) with a snubber circuit. Both

sections are integrated into a final combined system.

Touch sensing circuit

A capacitive touch sensor is used for the touch system. This sensor operates effectively even behind plastic or glass enclosures by detecting changes in capacitance, making it suitable for sleek, sealed interfaces. The TTP223 IC allows selection of various touch behaviours as defined in the datasheet. Fig. 3 shows IC TTP223 pinout details, and Fig. 4 presents the touch sensing circuit.

The TTP223-BA6 IC operates between 2.0V and 5.5V. It requires a sensing electrode for touch detection, which can be a piece of conductive material connected to the appropriate IC pin. The IC provides a digital output signal that changes state when a touch is detected.

The IC can be configured in four modes using a solder jumper (J1) between TOG (pin 6) and AHLB (pin 4):

- 1. Push mode (TOG = 0, AHLB = 0).** Output goes high when touched and low when released; suitable for momentary functions.
- 2. Toggle mode (TOG = 1, AHLB = 0).** Each touch toggles output between high and low; suitable for on/off switching.
- 3. Toggle mode with power-on state = 0 (TOG = 1, AHLB = 1).** Output starts low and toggles with each touch; suitable for default-off devices.
- 4. Toggle mode with power-on state = 1 (TOG = 1, AHLB = 1).** Output starts high and toggles with each touch; suitable for default-on devices.

Solid-state relay circuit

The SSR design utilises the BT136 SMD TRIAC IC, suitable for switching general-purpose motors, lights, and household appliances. The BT136 handles up to 600V RMS (blocking voltage) and 4A RMS (on-state current), making it ideal for 100-250V AC input and corresponding loads.